

AtlasResearch

Tools and Technology Used

Technology	Description
Python 3.11	Primary programming language for application development.
FastAPI	High-performance backend framework for APIs.
Uvicorn	ASGI server used to run the FastAPI application.
PostgreSQL 17	Relational database for structured data storage.
pgvector	Extension enabling vector storage and similarity search.
Ollama	Local runtime for executing open-source AI models.
Open-Source LLM Inference	Runs large language models locally without cloud APIs.
all-MiniLM-L6-v2 (Embedding Model)	Generates semantic vector embeddings from text.
sentence-transformers	Library for creating and managing text embeddings.
langchain-huggingface	Integration layer for Hugging Face embedding models.
Web Research Pipeline (DuckDuckGo + Trafilatura)	Performs web search and extracts clean article content.
Custom Multi-Agent Architecture	Organizes Gatherer, Synthesizer, and Critic agents.
Agent Orchestration	Coordinates agent execution and workflow.
WebSockets	Enables real-time communication with the frontend.
React	Frontend library for building the user interface.
Vite	Fast build tool and development server for React.
Tailwind CSS	Utility-first CSS framework for styling.
Pydantic	Data validation and schema management library.
python-dotenv	Loads environment variables from .env files.
psycopg2-binary	PostgreSQL database adapter for Python.
Git	Distributed version control system.
GitHub	Platform for source code hosting and collaboration.